WARAITIP LAOSANGPRATEEP

(+66) 62-169-9642 · waraitip.l26@gmail.com · LinkedIn · Portfolio
Thai national · Requires Employment Pass sponsorship · Available within 1 month

Application Specialist for Thermo Fisher SEM at LMS Instruments (Thailand), with intensive Talos TEM training and an M.Eng. in Medical Engineering (University of Auckland, Distinction). Support customers across PCB and forensic applications, from engineers to shopfloor staff, and explain technical results clearly to specialist and non-specialist audiences. Completed semiconductor process coursework (KAIST, Coursera) and contributed to a group business case (SSBC). Analyse data in Python, SQL, and Power BI. Seeking a Business Development or Product Management role in Singapore's semiconductor and advanced technology sector.

EXPERIENCE

Application Specialist – Thermo Fisher EM product line Oct 2026 – Present

LMS Instruments Co., Ltd. · Thailand

- Meet PCB-industry customers (KCE) and forensic customers (RPCA) to uncover their needs and technical challenges, and match them to the Thermo Fisher SEM that fits their application.
- Research and identify potential customers and organisations likely to be interested in the products, supporting business development for the SEM product line.
- Built a competitor landscape for Thailand covering gunshot residue (GSR) analysis and Amira/Avizo 3D reconstruction software: gathered and classified the data independently, then structured it into an Excel comparison ready for team use, using LLM tools to speed up setup.
- Provide SEM application support and explain technical results and instrument value to specialist and non-specialist audiences.

Research Intern – Nanomaterial Characterisation (SWCNTs) Jun 2023 – Jul 2023

Academia Sinica, Institute of Atomic and Molecular Sciences (IAMS-IIP), with Prof. Ching-Wei Lin · Taipei, Taiwan **Link**

- Characterised SWCNT defect populations by absorbance and fluorescence spectroscopy for cancer diagnostics.
- Designed a DOE-style experimental matrix and processed multi-dimensional spectral data in Origin Lab, contributing findings to peer reports.
- Operated experiments independently within days of arrival in a new country and team.

Customer Service & Operations Aug 2024 – Jan 2026

Auckland, New Zealand

- Part-time customer-facing hospitality and retail roles (Lily's Collection, Blue Elephant, Mi&Chi, Khao San Eatery, The Coffee Club) while completing a research-intensive degree. Resolved day-to-day operational and customer issues independently across multiple fast-paced venues, requiring minimal supervision.

EDUCATION

Master of Medical Engineering – University of Auckland, NZ Mar 2025 – Jun 2026

GPA 7.00 / 9.00 · Graduated with Distinction (Referees available upon request)

- **Thesis:** mechanical characterisation of neonatal lamb growth plate by microindentation, mapping stiffness and hardness across four structural zones. Ran the full pipeline from cryogenic tissue preparation to data acquisition, outlier analysis and statistical testing.
- **Group project, Business Case:** co-developed a Single Stage Business Case for a regional EV charging programme (Auckland), covering strategic, financial and operational feasibility, multi-criteria analysis, delivery models, funding pathways and risk.
- **Project, Machine Learning:** built and cross-validated SVM models in Python to classify meat types from spectral data.

- **Coursework:** Advanced Functional Materials · Battery & Superconductor · Semiconductor & Materials Science · Medical Device & Technology Development · Advanced Imaging Technologies (OCT, MRI) · Machine Learning for Biomedical Applications · Advanced Drug Disposition & Kinetics · Engineering Project Management · Medical Device Development & regulation · Project Business Case

B.Eng. Chemical Engineering – Thammasat University (SIIT), Thailand Aug 2020 – May 2024
International Programme · GPA 3.22 / 4.00

- Senior project: computational study of inclusion complexes using molecular docking (AutoDock) and quantum chemical modelling (Gaussian); presented as a poster at PACCON 2024.
- **Coursework:** Chemical Reaction Engineering · Process Design · Plant Design · Thermodynamics · Heat & Mass Transfer · Fluid Mechanics · Environmental Engineering · Wastewater Treatment · Safety in Chemical Operations · Analytical and Instrumental

SKILLS

- **Thermo Fisher Scientific Platforms**
  Thermo Scientific Axia SEM · Phenom SEM · Talos TEM (intensive training) · Avizo 3D image analysis (trained)
- **Semiconductor & Advanced Technology**
  Semiconductor manufacturing and wafer fab fundamentals · Process control · Defect inspection and failure analysis concepts · Yield improvement · PCB and electronics inspection· Thin films and nanostructures · SEM/TEM for materials and failure analysis Electrode · separator · material characterisation · structure-property relationships
- **Hands-on Technical Expertise: Nano/Microstructure Materials**
  SEM and TEM imaging · Microindentation · DIC microscopy · Absorbance and fluorescence spectroscopy · Mechanical property testing · Nanomaterial characterisation · Technical report writing
- **Business & Commercial**
  Business case development (SSBC group project) · Financial and operational feasibility analysis · Multi-criteria and risk analysis · Customer and stakeholder communication · Technical presentations · Cross-functional project management · Waste Reduction and Recycling Technologies · Product Stewardship · Engineering Management
- **Data, Machine Learning & Process**
  Python (pandas, NumPy, scikit-learn, matplotlib) · SQL · Power BI · Statistical analysis · Supervised Machine learning (SVM classification, cross-validation) · Design of Experiments (DOE) · Root cause analysis · FMEA
- **Medical devices, Pharma, and life sciences**
  Pharmacokinetics and drug disposition · ADME · PK modelling · Drug-material interaction modelling (molecular docking) · GMP Design verification and validation · ISO 13485 · ISO 14971 · IEC 60601-1 · IEC 62366 · Material and coating inspection, Failure analysis

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Data Analysis with SQL, Excel & Power BI Specialisation; Google Data Analysis with Python Specialisation (Coursera)
- KAIST Introduction to Semiconductor Process 1 & 2 (Coursera)
- Talos TEM and Avizo 3D training (STREC, Chulalongkorn University, LMS Thailand, Thermo Fisher)
- Poster presentation, PACCON 2024; Academia Sinica IAMS-IIP internship certificate

LANGUAGES

Thai (native) · English (IELTS 6.5) · Korean (elementary; King Sejong Institute Level 1B, TOPIK I target Dec 2026)